Yashwardhan Panchal

yashpanchal1326@gmail.com | linkedin.com/in/yashpanchal | github.com/yashpanchal1012 | Portfolio:yashpanchal1012.github.io.

Education

University of Minnesota, Twin Cities — Minneapolis, MN
Bachelor of Science in Computer Science — GPA: 3.8 / 4.0 (Dean's List 2023–2026) — *Sep 2023 – Dec 2026*

Relevant Coursework: Robotics (TA), Computer Vision, Applied Machine Learning, Natural Language Processing

Technical Skills

Programming Languages: Python, C, C++, Java, MATLAB

Robotics & Machine Learning: ROS2, OpenCV, TwinCAT, MediaPipe, TensorFlow, PyTorch, SciKit-Learn, Matplotlib, Seaborn, NumPy, Pandas, Gradio, ONNX, SLAM,

Web Development: HTML, CSS, Bootstrap, Django, JavaScript, SQL, React, Next.js, Tailwind

Other: Git, LaTeX, Windows, Unix/Linux, Docker, EtherCAT

Experience

Automation Engineering Intern — June 2026 – Aug 2026
Beckhoff Automation — *Savage, MN*

- Developed a gesture-controlled TwinCAT Vision system to manipulate XTS linear movers in real time.
- Assisted application engineers in technical meetings and managed client project data within CRM.
- Partnered with cross-functional engineering teams to deliver **4+** technical solution proposals and proof-of-concept demos for enterprise clients.

Depot Assistant — Jan 2024 – Present
Electrical and Computer Engineering Department, UMN — *Minneapolis, MN*

- Maintained lab equipment for multiple Electrical and Computer Engineering courses.
- Assembled **300+** lab kits and performed basic soldering and repairs.
- Communicated daily with faculty and engineering students to advise on part selection, inventory logistics, and lab kit assembly.

Undergraduate Teaching Assistant — Sep 2024 – Present
Computer Science & Engineering Department, UMN — *Minneapolis, MN*

- Collaborated with course professors and TA teams to align weekly curriculum, lead labs and exam prep for **20+** students.
- Held office hours and graded assignments with detailed feedback.

Projects

Vision-Operated Robotic Arm
Robotics & Computer Vision — *ROS2, MediaPipe, Gazebo, OpenCV, Python*

- Led a 3-person team to build a real-time ROS2 pipeline translating webcam-tracked human motion into robotic joint commands
- Implemented 30 Hz MediaPipe skeletal tracking, computing joint angles via vector mathematics
- Simulated physics-enabled URDF models in Gazebo simulation, tuning motion profiles for stable trajectory execution
- Connected human movement data to robotic control logic for responsive, real-time motion translation.

Semantic Image Search
Machine Learning & Computer Vision — *PyTorch, CLIP, Gradio, Python*

- Built a multimodal image retrieval tool that ranks a local image collection against natural-language or image queries.
- Encoded the searchable image cache with OpenAI CLIP and normalized embeddings for cosine-similarity retrieval.
- Supported both text prompts and uploaded reference images through a Gradio interface.
- Added configurable top-k results with similarity scores so users can inspect and compare ranked matches.

Traffic Sign Recognition
Machine Learning & Computer Vision — *PyTorch, DINOv2, Qwen3-VL, Hugging Face, Python*

- Developed a GTSRB traffic sign perception pipeline, achieving 76.6% accuracy across 43 classes
- Built a 3-layer PyTorch MLP classifier on top of a frozen DINOv2 vision transformer backbone
- Evaluated a zero-shot Qwen3-VL baseline to analyze real-time latency and reliability trade-offs